

Examples of Macro Usage

Generalized Tensors (K, L, $\tilde{A}$)

Formula (58) involves:

$$\begin{aligned} K_{(\mu\nu)m}^{(\kappa\lambda)} &= (\kappa\lambda+)^{(1,6)}_{;} \delta(l, m) - (\kappa\lambda+)^{(1,6)}_{(\mu\nu)(-)m} \\ L_{(\mu\nu)}^{(\kappa\lambda)} &\quad (\text{Example for L}) \\ \tilde{A}_{(\mu\nu)}^{(\kappa\lambda)} &\quad (\text{Example for } \tilde{A}) \end{aligned}$$

The term $Q_{(\mu\nu)}^{(\kappa\lambda)}$ often appears with an additional subscript like $Q_{e(\mu\nu)}^{(\kappa\lambda)}$. A more general macro for symbols with two pairs of indices could be:

```
\newcommand{\HeimGenTensor}[3]{#1^{(#2)}_{(#3)}}
Usage:
\HeimGenTensor{Q_e}{\kappa\lambda}{\mu\nu}
gives  $Q_{e(\mu\nu)}^{(\kappa\lambda)}$ 
(adjusting for actual notation).
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Square Brackets

From formula (20): $[kl]$ gives $[kl]$
 From formula (56a): $[] = [] + [[]]^x$ gives $[] = [] + []^x$
 From formula (57), a term like $[\kappa\lambda]$ or $[\kappa\lambda+]$

Quantized C Symbol (Formula 3)

$$C_{(p)km}^{(p)} = \lambda_{(p)}(k, m)\phi_{km}^{(p)} \text{ gives } C_{(p)km}^{(p)} = \lambda_{(p)}(k, m)\phi_{km}^{(p)}$$

Connecting to Christoffel Symbols and Quantization

The standard Christoffel symbol Γ_{km}^l (formula 1b) has one upper and two lower indices. The quantized version in formula (3), $C_{(p)km}^{(p)}$, introduces a qualifier p which seems to operate on both levels. The Heim tensor $K_{(\mu\nu)m}^{(\kappa\lambda)}$ (formula 58) is a more complex analogue.

- The superscripted pair $(\kappa\lambda)$ can be seen as analogous to the single upper index l in Γ_{km}^l .
- The subscripted pair $(\mu\nu)$ can be seen as analogous to the lower indices k, m in Γ_{km}^l .
- The additional index m in $K_{(\mu\nu)m}^{(\kappa\lambda)}$ and the qualifiers like ‘+’, ‘(1,6)’, ‘(-)’ indicate a more generalized or structured connection, likely arising from the specific geometry or algebra of Heim’s theory.

The RHS of the definition of $K_{(\mu\nu)m}^{(\kappa\lambda)}$ shows how it’s constructed from more fundamental parenthesized terms: $(\kappa\lambda+)^{(1,6)}_{;}$ and $(\kappa\lambda+)^{(1,6)}_{(\mu\nu)(-)m}$. These terms themselves encapsulate pairs of indices $(\kappa\lambda)$, a sign +, and then further operational superscripts/subscripts like (1,6), ;, or the complex $(\mu\nu)(-)m$. This operational aspect is where the ”matrix-like” nature comes in – they are not just passive indices but define transformations or components within a larger structure.

The square brackets ‘[]’ appear to be fundamental operators or quantities in this formalism. For example, in (57):

$$\tilde{\Gamma} = \sum_e \left([\mu\nu] + \text{sp}^2 Q_e^{(\mu\nu)}(\kappa\lambda) \times [\kappa\lambda] \right) \times [\kappa\lambda+]$$

Empty Index Elements (Formula 7)